

CONCEPT MAP

CSA6503 – Concept Map

Artificial Intelligence, Machine Learning, Deep Learning, Generative AI & Large Language Models



LAYER 1 · BROADEST FIELD

Artificial Intelligence (AI)

The broad field of creating systems capable of performing tasks requiring human intelligence.

EXAMPLE Self-driving cars (Tesla)

AI → ML: Machine Learning is a subset of Artificial Intelligence.



LAYER 2 · SUBSET OF AI

Machine Learning (ML)

Algorithms that learn patterns from data without being explicitly programmed.

EXAMPLE Netflix Recommendation System

ML → DL: Deep Learning is a subset of Machine Learning that uses multi-layer neural networks.



LAYER 3 · SUBSET OF ML

Deep Learning (DL)

A branch of Machine Learning using neural networks with many hidden layers.

EXAMPLE Face Recognition (Apple Face ID)

DL → **Generative AI**: Generative AI uses deep learning models to generate new content.



LAYER 4 · APPLIES DL

Generative AI

AI capable of generating new text, images, music, code, and other content.

EXAMPLE Adobe Firefly

Generative AI → LLMs: Large Language Models are a type of Generative AI specialized for understanding and generating human language.



LAYER 5 · SPECIALIZED GENERATIVE AI

Large Language Models (LLMs)

Large neural networks trained on massive text datasets to understand and generate human language.

EXAMPLE ChatGPT

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Generative AI and Large Scale Models